Lab 03

Introducing Java Applications II

Problem Solving Exercises

1. Write a Java program to print the sum of all numbers from 1 to 100.

2. Write a Java program to swap two variables.

3. Write a Java program and compute the sum of the digits of an integer.

   Sample Output:

   Input an integer: **25**

   ```
   The sum of the digits is: 7
   ```

4. Write a Java program to print numbers between 1 to 100 which are divisible by 3, 5 and by both.

   Sample Output:

   ```
   Divided by 3:
   3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51,
   54, 57, 60, 63, 66, 69, 72, 75, 78, 81, 84, 87, 90, 93, 96, 99,

   Divided by 5:
   5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85,
   90, 95,

   Divided by 3 & 5:
   15, 30, 45, 60, 75, 90,
   ```

5. Write a Java program that accepts four integers from the user and prints equal if all four are equal, and not equal otherwise

   Sample Output:

   Enter the first number: **25**
   Enter the second number: **37**
   Enter the third number: **45**
   Enter the fourth number: **23**
   Numbers are not equal!

6. Write a Java program to request the user to type the time in seconds and display it in the standard format of `hours:minutes:seconds`.

```
Time in seconds: 3725
--------------------
Time is . . .   1:2:5

Press any key to continue . . .
```

7. Write a program that prompts for two numbers. Print the sum, difference, product, and quotient of those numbers as shown in the example output:

Sample Output:

What is the first number? **10**
What is the second number? **5**
10 + 5 = 15
10 – 5 = 5
10 * 5 = 50
10 / 5 = 2

Constraints

Values received from users will be strings. Ensure that you convert these values to numbers before doing the math.

8. Write a program to evenly divide pizzas. Prompt for the number of people, the number of pizzas, and the number of slices per pizza. Ensure that the number of pieces comes out even. Display the number of pieces of pizza each person should get. If there are leftovers, show the number of leftover pieces.

Sample Output:

How many people? **8**
How many pizzas do you have? **2**
How many slices per pizza? **10**
8 people with 2 pizzas...
Each person gets 2 slices of pizza.
There are 4 leftover slices.

9. Write a program that generates multiplication tables for the numbers 0 through 12.

Sample Output:

0 X 0 = 0
0 X 1 = 0
...
12 x 11 = 132
12 x 12 = 144

Constraints

Use a nested loop to complete this program.

10. Write a Java program that takes two points from the user (x1, y1) & (x2, y2) then calculates the distance between them, given this formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Sample Output:

Enter 4 integer values for (x1, y1) & (x2, y2) respectively:
4
2
7
4

Distance between (4, 2) & (7, 4) is 3.61

11. Write a Java program that keeps accepting numbers from the user until a negative value is entered. The program then calculates number of entered values, their sum & average. Format your output to show the average to 2 decimal places.

Sample Output:

enter positive values, or –1 to quit:
9
enter positive values, or –1 to quit:
5
enter positive values, or –1 to quit:
6
enter positive values, or –1 to quit:
3
enter positive values, or –1 to quit:
1
enter positive values, or –1 to quit:
–1
You entered 5 numbers, sum is 24, average is 4.80